

SEPHORA COSMETIC REVIEWS: AI-Powered Review Rating System

1. Dataset Used : Sephora Products and Skincare Reviews

This dataset contains: information about all beauty products (over 8,000) from the Sephora online store, including product and brand names, prices, ingredients, ratings, and all features. user reviews (about 1 million on over 2,000 products) of all products from the Skincare category, including user appearances, and review ratings by other users.

1.1 Objective

Build an automated machine learning system that predicts the star rating (1-5 stars) of a cosmetic product review based purely on the text content. This is a supervised, multi-class text classification problem.

1.2 Dataset

Loaded the Sephora Cosmetic reviews dataset using Pandas. The dataset initially contained 602,130 rows and 19 columns.

Inspected the dataset using `head()`, `shape`, `columns`, `info()`, and missing-value analysis. The 19 raw columns:

Column	Data Type	Description
Unnamed: 0	int64	Auto-index (dropped)
author_id	object	Unique reviewer identifier (dropped)
rating	int64	Star rating 1-5 — TARGET VARIABLE
is_recommended	float64	1.0=recommended, 0.0=not (117,486 nulls)
helpfulness	float64	Helpfulness ratio score (331,832 nulls)

total_feedback_count	int64	Total helpful votes received
total_neg_feedback_count	int64	Negative helpfulness votes (dropped)
total_pos_feedback_count	int64	Positive helpfulness votes (dropped)
submission_time	object	Review date YYYY-MM-DD (dropped)
review_text	object	Full review text — PRIMARY INPUT
review_title	object	Short review title (167,011 nulls)
skin_tone	object	Self-reported skin tone (106,056 nulls)
eye_color	object	Self-reported eye color (dropped)
skin_type	object	Self-reported skin type (74,683 nulls)
hair_color	object	Self-reported hair color (dropped)
product_id	object	Product identifier (dropped)
product_name	object	Full product name
brand_name	object	Brand name
price_usd	float64	Product price USD (dropped)

2. Libraries and Environment

2.1 Imports Used

Library	Module / Component	Purpose
pandas	pandas	Data loading, cleaning, grouping, feature engineering
nltk	nltk.corpus.stopwords	English stopwords list
nltk	nltk.stem.WordNetLemmatizer	Word lemmatization (root form extraction)
nltk	nltk.download()	Downloads stopwords,

		wordnet, omw-1.4 data
re	re (standard library)	Regular expressions for URL/HTML/special char removal
string	string (standard library)	string.punctuation for punctuation stripping
matplotlib	matplotlib.pyplot	All chart rendering
seaborn	seaborn	Statistical plots: countplot, histplot, boxplot, barplot
sklearn	model_selection.train_test_split	Stratified 80/20 split
sklearn	feature_extraction.text.TfidfVectorizer	TF-IDF vectorization

2.2 NLTK Downloads

```
import nltk
nltk.download('stopwords') # English stopword list
nltk.download('wordnet') # Lemmatization vocabulary
nltk.download('omw-1.4') # Open Multilingual Wordnet (required by wordnet)
```

3. Data Loading and Initial Inspection

df.head(5)

Confirmed column names, data types, and that review_text contains natural English. Showed is_recommended, helpfulness, review_title, skin_tone already have NaN in the first rows.

df.shape -> (602130, 19)

df.columns.tolist()

Listed all 19 column names. Confirmed review_text and rating are present.

df.dtypes

Confirmed rating is int64 (no conversion needed), 5 int columns, 3 float columns, 11 object/string columns.

df.isnull().sum() - Missing Values Audit

Column	Null Count	% Missing	Action
review_text	999	0.17%	Dropped rows (critical)
review_title	167,011	27.7%	Column kept
is_recommended	117,486	19.5%	Column kept
helpfulness	331,832	55.1%	Column kept
skin_tone	106,056	17.6%	Column kept
eye_color	138,488	23.0%	Column dropped
skin_type	74,683	12.4%	Column kept
hair_color	141,081	23.4%	Column dropped
All other columns	0	0%	No action

Initial Rating Distribution

Rating	Count	Percentage
★ 1	32,903	5.47%
★ 2	29,253	4.86%
★ 3	44,018	7.31%
★ 4	106,956	17.76%
★ 5	389,000	64.60%

4. Data Cleaning

Identified missing values across all columns. Removed rows with missing review_text values (999 records).

Verified that the rating column contained no missing values and only valid ratings from 1 to 5.

Missing Values:

Unnamed: 0	0
author_id	0
rating	0

is_recommended	117486
helpfulness	331832
total_feedback_count	0
total_neg_feedback_count	0
total_pos_feedback_count	0
submission_time	0
review_text	999
review_title	167011
skin_tone	106056
eye_color	138488
skin_type	74683
hair_color	141081
product_id	0
product_name	0
brand_name	0
price_usd	0

4.1 Remove Null Review Text

999 rows where review_text was NaN were dropped - these have no input for the model.

```
df = df.dropna(subset=['review_text'])  
# 602,130 -> 601,131 rows
```

A follow-up check confirmed 0 empty string reviews after whitespace stripping.

4.2 Verify Rating Values

```
df['rating'].unique() -> [5 1 4 2 3]  
# All values are exactly integers 1-5, no out-of-range values
```

4.3 Duplicate Analysis

Checked for exact duplicate records and removed them where applicable.
Analyzed conflicting reviews (same review text with different ratings). Only about 100 conflicting review texts were found.
These were retained because they represent real-world variation in user opinions and formed a negligible portion of the dataset.

Duplicate review texts (same text, different rows)

```
df['review_text'].duplicated().sum()  
# Duplicate Review Texts: 92,664
```

92,664 rows share text with at least one other row. Not dropped because the same review text may legitimately appear for different products.

Conflicting reviews (same text, different ratings)

```
conflicting = df.groupby('review_text')['rating'].nunique()  
conflicting = conflicting[conflicting > 1]
```

Duplicate review text analysis identified 100 conflicting review texts associated with multiple rating values. Since these represented less than 0.02% of the dataset and may reflect genuine differences in user perception, they were retained in the final dataset.

4.4 Column Selection

From 19 original columns, 9 were kept:

```
# Shape: (601,131, 9)
```

Status	Columns
Kept	review_text, rating, review_title, is_recommended, helpfulness, skin_type, skin_tone, product_name, brand_name
Dropped	Unnamed:0, author_id, total_feedback_count, total_neg_feedback_count, total_pos_feedback_count, submission_time, eye_color, hair_color, price_usd

5. Text Preprocessing

5.1 NLTK Setup

```
from nltk.corpus import stopwords  
from nltk.stem import WordNetLemmatizer
```

```
stop_words = set(stopwords.words('english')) # 179 stopwords
lemmatizer = WordNetLemmatizer()
```

5.2 The preprocess_text() Function

A single reusable function applies 8 sequential cleaning operations to each review:

```
def preprocess_text(text):
    text = str(text).lower()                # 1. Lowercase
    text = re.sub(r'http\S+|www\S+|https\S+', '', text) # 2. Remove URLs
    text = re.sub(r'<.*?>', '', text)        # 3. Strip HTML
    text = text.translate(                  # 4. Remove punctuation
        str.maketrans("", "", string.punctuation))
    text = re.sub(r'^a-zA-Z0-9\s]', '', text)    # 5. Special chars
    text = re.sub(r'\s+', ' ', text).strip()      # 6. Collapse spaces
    words = text.split()
    words = [lemmatizer.lemmatize(word)          # 8. Lemmatize
              for word in words
              if word not in stop_words]         # 7. Remove stopwords
    return ' '.join(words)
```

Stage	Operation	Before	After
1	Lowercase	'Great Product!'	'great product!'
2	Remove URLs	'Buy at https://sephora.com'	'Buy at '
3	Strip HTML	'Rated 5 stars'	'Rated 5 stars'
4	Remove punctuation	'great, perfect!'	'great perfect'
5	Special characters	'100% pure'	'100 pure'
6	Collapse whitespace	'great skin care'	'great skin care'

7	Remove stopwords	['i','really','love','this']	['really','love']
8	Lemmatize	['hydrating','loved']	['hydrate','love']

Stopwords :

['a', 'about', 'above', 'after', 'again', 'against', 'ain', 'all', 'am', 'an', 'and', 'any', 'are', 'aren', 'aren't', 'as', 'at', 'be', 'because', 'been', 'before', 'being', 'below', 'between', 'both', 'but', 'by', 'can', 'couldn', 'couldn't', 'd', 'did', 'didn', 'didn't', 'do', 'does', 'doesn', 'doesn't', 'doing', 'don', 'don't', 'down', 'during', 'each', 'few', 'for', 'from', 'further', 'had', 'hadn', 'hadn't', 'has', 'hasn', 'hasn't', 'have', 'haven', 'haven't', 'having', 'he', 'he'd', 'he'll', 'he's', 'her', 'here', 'hers', 'herself', 'him', 'himself', 'his', 'how', 'i', 'i'd', 'i'll', 'i'm', 'i've', 'if', 'in', 'into', 'is', 'isn', 'isn't', 'it', 'it'd', 'it'll', 'it's', 'its', 'itself', 'just', 'll', 'm', 'ma', 'me', 'mightn', 'mightn't', 'more', 'most', 'mustn', 'mustn't', 'my', 'myself', 'needn', 'needn't', 'no', 'nor', 'not', 'now', 'o', 'of', 'off', 'on', 'once', 'only', 'or', 'other', 'our', 'ours', 'ourselves', 'out', 'over', 'own', 're', 's', 'same', 'shan', 'shan't', 'she', 'she'd', 'she'll', 'she's', 'should', 'should've', 'shouldn', 'shouldn't', 'so', 'some', 'such', 't', 'than', 'that', 'that'll', 'the', 'their', 'theirs', 'them', 'themselves', 'then', 'there', 'these', 'they', 'they'd', 'they'll', 'they're', 'they've', 'this', 'those', 'through', 'to', 'to o', 'under', 'until', 'up', 've', 'very', 'was', 'wasn', 'wasn't', 'we', 'we'd', 'we'll', 'we're', 'we've', 'were', 'weren', 'weren't', 'what', 'when', 'where', 'which', 'while', 'who', 'whom', 'why', 'will', 'with', 'won', 'won't', 'wouldn', 'wouldn't', 'y', 'you', 'you'd', 'you'll', 'you're', 'you've', 'your', 'yours', 'yourself', 'yourselves']

5.3 Before / After Examples from Notebook

ORIGINAL:

I use this with the Nudestix "Citrus Clean Balm & Make-Up Melt" to double cleanse and it has completely changed my skin (for the better). The make-up melt is oil based and removes all of your makeup super easily. I follow-up with this water based cleanser, and I also use this just by itself when I'm not wearing make-up. It leaves the skin gently cleansed, but without stripping the skin. 10/10 recommend combining with the make-up melt. It's perfection!

CLEANED:

use nudestix citrus clean balm makeup melt double cleanse completely changed skin better makeup melt oil based remove makeup super easily followup water based cle

anser also use im wearing makeup leaf skin gently cleansed without stripping skin 10
10 recommend combining makeup melt perfection

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ORIGINAL:

I bought this lip mask after reading the reviews and the hype. Unfortunately, it did not meet my expectations as vaseline petroleum jelly works way better for me.

CLEANED:

bought lip mask reading review hype unfortunately meet expectation vaseline petroleum jelly work way better

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ORIGINAL:

My review title says it all! I get so excited to get into bed and apply this lip mask. I do see a difference because I suffer from dry cracked lips. I drink a lot of water and apply lip balm daily but nothing helped until I started using this. untiluntistafted usinf this.

CLEANED:

review title say get excited get bed apply lip mask see difference suffer dry cracked lip drink lot water apply lip balm daily nothing helped started using untiluntistafted usinf

6. Review Length Filtering

6.1 Word Count Statistics

```
df['word_count'] = df['clean_review'].str.split().str.len()
```

Statistic	Value	Interpretation
count	601,044	Reviews after null removal
mean	30.14	Average: ~30 cleaned words per review

std	21.73	High variance in review length
min	1	Shortest reviews: single word
25%	16	Bottom quarter: 16 words or fewer
50% (median)	25	Half of reviews have 25 words or fewer
75%	38	Top quarter: 38+ words
max	525	Longest review outlier

6.2 Filter Applied

```
df = df[(df['word_count'] >= 3) & (df['word_count'] <= 300)]
# Shape after: (600,342, 16)
```

Threshold	Value	Rationale
Minimum	3 words	Reviews with 1-2 cleaned words carry no meaningful sentiment signal
Maximum	300 words	Very long reviews are outliers (<0.3% of data) and skew TF-IDF vectors

7. Feature Engineering

Five numerical features were derived from the review text and stored as new DataFrame columns:

```
df['word_count'] = df['clean_review'].str.split().str.len()
df['char_count'] = df['clean_review'].str.len()
df['review_length'] = df['clean_review'].str.len() # same as char_count
df['avg_word_length'] = (df['char_count'] / df['word_count']).round(2)
df['exclamation_count'] = df['review_text'].astype(str).str.count('!')
df['question_count'] = df['review_text'].astype(str).str.count(r'\?')
```

Feature	Source	Mean	Std	Min	Max
word_count	clean_review	30.16	21.65	3	295
char_count	clean_review	195.63	142.62	8	1,949
review_length	clean_review	195.63	142.62	8	1,949

avg_word_length	clean_review	6.46	0.57	2.00	18.38
exclamation_count	review_text	--	--	0	varies
question_count	review_text	--	--	0	varies

8. Class Distribution Analysis

After all cleaning and filtering, the final class distribution before balancing:

Rating	Count	Percentage	Observation
★ 1	32,789	5.46%	Negative reviews - minority class
★ 2	29,195	4.86%	Below average - rarest class
★ 3	43,950	7.32%	Neutral reviews
★ 4	106,747	17.78%	Good reviews
★ 5	387,661	64.57%	Dominant class (nearly 2 in 3 reviews)
Total	600,342	100%	--

9. Data Visualization

Chart 1 - Review Count per Rating (Bar Chart)

`sns.countplot` showing raw number of reviews per class. Clearly shows the 5-star dominance and motivates class balancing.

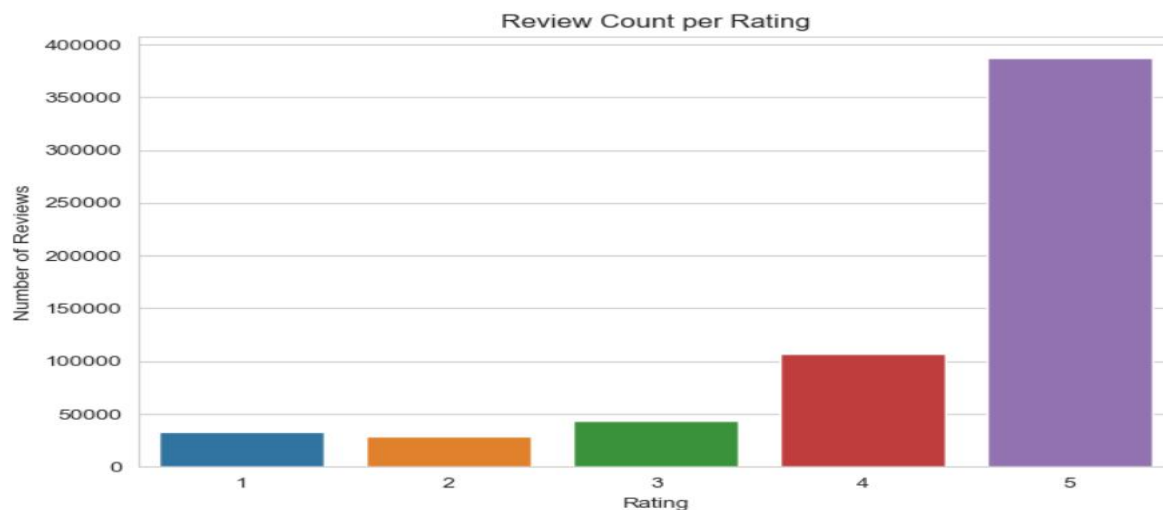


Chart 2 - Word Count Distribution by Rating (Histogram)

Overlapping histplot with alpha=0.4, one per rating class. All classes right-skewed. 1-star and 3-star reviews tend to be slightly longer - dissatisfied or neutral customers write more detailed explanations.

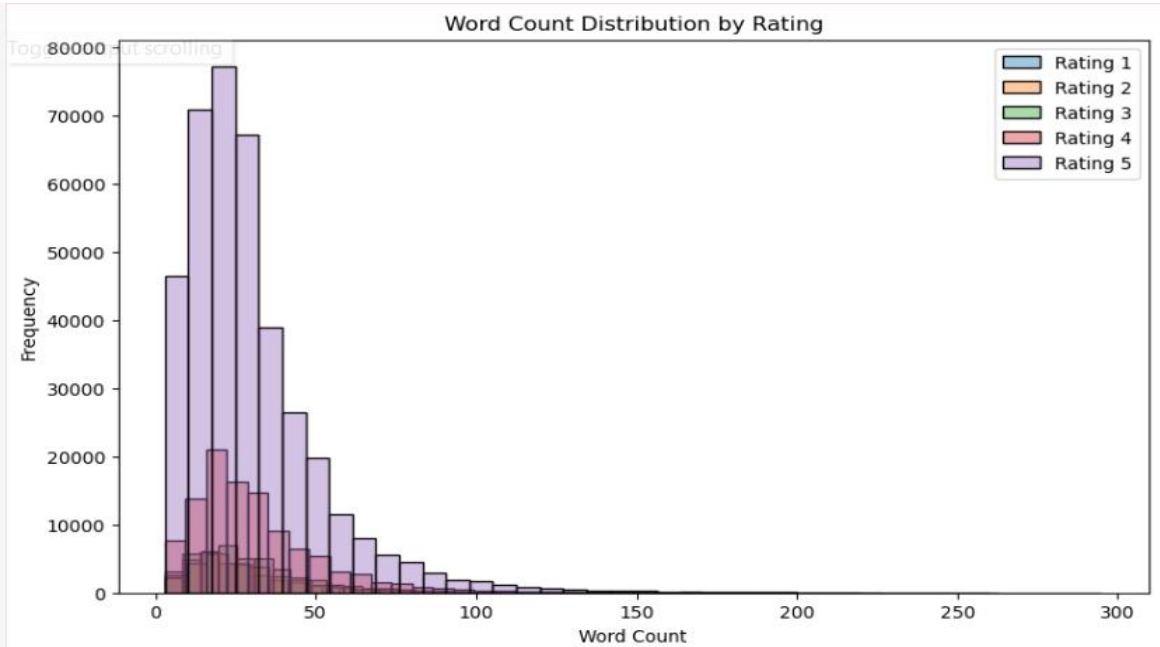


Chart 3 -

Word Count per Rating (Box Plot)

sns.boxplot comparing spread and median word count across all 5 classes. Median similar across classes (~25-32 words). Rating 3 has widest IQR - most variable length. All classes show outliers above 100 words.

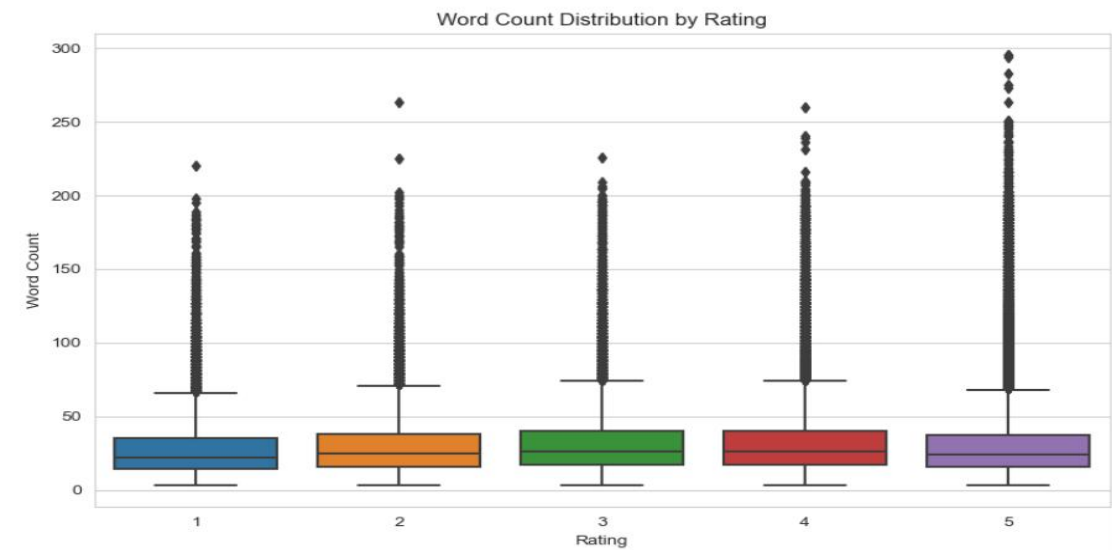


Chart 4 – Review Length Distribution by Rating (Violin Plot)

A violin plot was used to visualize the density and distribution of review lengths for each rating category.

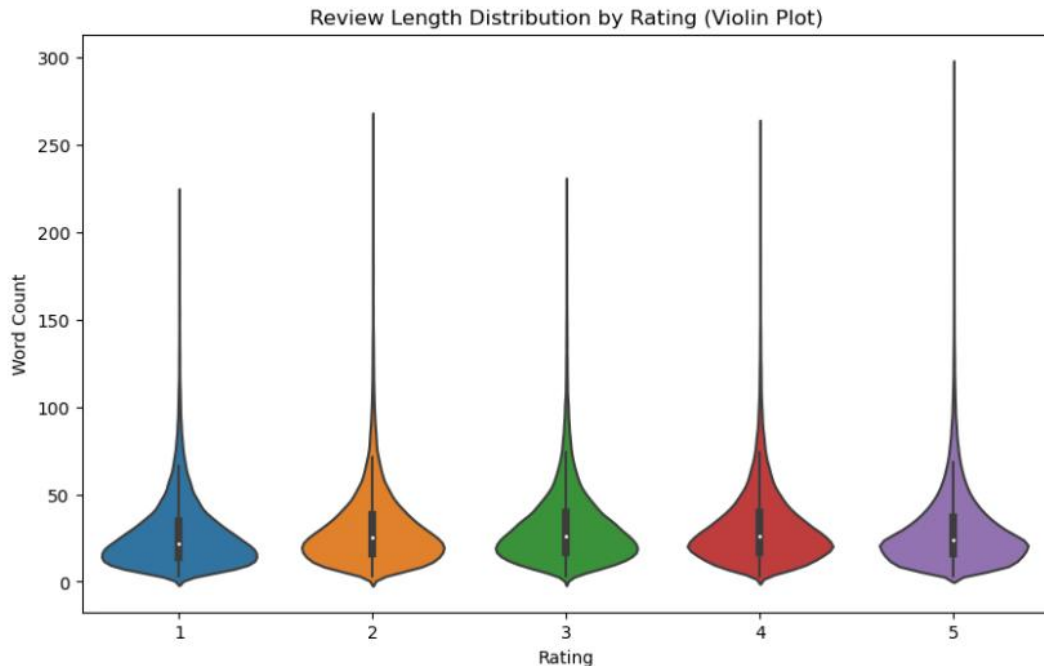


Chart 4 - Sample Reviews per Rating (Text)

5 raw reviews per class printed as a qualitative check (not a chart). Key vocabulary patterns observed:

- 1-star: 'ruined my skin', 'broke out', 'cystic acne', 'returning it'
- 2-star: nuanced criticism - 'not bad but', 'overhyped', 'packaging horrible'
- 3-star: balanced - 'it's okay', 'works for some but not me', 'decent'
- 4-star: mostly positive with one specific complaint about price or scent
- 5-star: superlatives - 'holy grail', 'game changer', 'absolutely love it'

RATING 1

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Review 1

Not for everyone. I got this part of Influenster voxbox for testing purposes. I put it on after cleansing and it tingled for a bit. After I was done, it got so much worse I had to wash it off. Patch test your inner elbow first.

Review 2

I used this product religiously for 4 weeks straight! It ruined my skin, made my pores huge and I broke out really bad! I've been trying to get my skin healthy again but it has not been easy! Be careful using this product! I had to return it.

Review 3

Made my skin super smooth at first. I used it a couple of days ago. Noticed some bumps on the left side of my face, then they spread EVERYWHERE. Is anyone else having this problem. My face is covered in these tiny, hard, skin colored bumps. When I say my face is covered, IT IS COVERED. I don't know what it is. I'm not blaming the product at all, my skin just doesn't like it. Darn you, skin!!

Review 4

I wanted to love this mask, but it sat greasy on my skin and gave me cystic acne which, after never having even so much as *regular* acne (sorry but it's true), was extremely disturbing and disappointing to me. I'm not sure why my skin reacted the way it did, but I ended up returning it.

Review 5

I wanted to like this product because it came highly recommended, but I quickly found that it was similar in consistency to petroleum jelly (like Vaseline) and didn't really do anything remarkable for my face. I felt like to properly wash it off my face after applying, it required me to tug at my skin, and nearly always left my pores feeling clogged and dry afterwards. Usually I have the opposite problem with products, as I feel like they strip the oil from my face and make my skin too tight and dry. After a wash, my face would feel far more textured by the end of the day. I really disliked this. To make matters worse, when I finished the bottle, the product had started to mold inside the bottle and around the cap. I won't be purchasing from Inkey list again after this one.

RATING 2

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Review 1

If you have never used argan oil before, Josie's might seem good. But compared to the argan oil I have been using, Josie's argan oil seems diluted, scentless (lacks the nutty natural scent, but maybe it is filtered differently) and too expensive.

Review 2

My skin is combo-oily, with an oily t-zone and dryness between my eyes, on my cheeks, and around my mouth. This didn't make me oilier or break out, but it did not take care of my dry spots, so back it went.

Review 3

I agree with so many of the reviewers of this product! The face wash itself is amazing but the packaging is horrible! Also, the bottle seems 1/2 full of product instead of there being a generous amount in there. The Tatcha emblem broke off upon first use when I squeezed the bottle. There needs to be a pump! I own almost every Tatcha product and this one doesn't hold the standard of quality that the rest do!

Review 4

I bought this product after reading all the good reviews hoping that it would be amazing and help clear some clogged pores. I have not noticed ANY difference at all in the 3 weeks i have been using this. I am really disappointed because i was sooo excited to get this, but it was a waste of money.

Review 5

I really wanted to love this product but unfortunately I cannot get past the scent!!! It's TERRIBLE! It smells like something along the lines of paint.... the mist itself came out evenly, and the finish on my face was nice but this smell! Can't get past it unfortunately.

RATING 3

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Review 1

This sunscreen leaves a metallic cast on your skin which looks awful if you're going barefaced. However, to be fair, it's advertised as a makeup primer. I was looking for something glowy and dewy for my dry skin, but this makes me look like a robot. I think you may get away with it if you are very, very fair. Otherwise you have to wear this under full coverage makeup. Three stars because it's advertised as a primer, but it did not work for me alone or with tinted moisturizer. How it's utilized I think explains the wildly differing reviews

Review 2

Save your money and purchase the Laneige lip mask instead. I really noticed no difference using the product and don't like the consistency as much either.

Review 3

I was asked to give an honest review of this via influenster and it honestly couldn't have come at a better time. In the winter I have to use heavier moisturizer on my face because of the drier air. Well this went on thick and went on smooth and seemed like it would be moisturizing enough but instead gave me dry patches all over. Not a fan and just wasn't for me.

Review 4

This didn't do much for me. I like a cleanser that is effective & rinses off easily. This doesn't do the latter. My sink splashes water all over the place so I prefer getting done in a hurry. This is just a preference thing; it's not a bad product.

Review 5

BRING BACK OLD FORMULA ;(glad I stocked up on the old one. Package is better.

RATING 4

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Review 1

This was a 100 point perk a few months ago, so I thought I'd try it out. Oh. My. God. do I love it! It's so nice to have around when my skin decides to break out every day! As soon as I notice a pimple forming, I'll dab some of this on at night, go to bed, wake up, wash my face and it's like the pimple was never there! The smell is a little off-putting at first, but you get over/use to it and BOY will this

product work miracles on your face. HIGHLY recommended for anyone that gets hormonal acne or pop-up pimples every now and then!

Review 2

It takes a lot for me to find something that I love enough to replace an item I'm already happy with. This is it. I made a purchase last week and chose this for a free sample. I feel like it's hard to find a face wash that doesn't strip your skin and this truly does not, and my face is already pretty dry on a normal basis. I got out of the shower after using this and my skin felt so smooth and supple to the point where I forgot to put on my face cream. There is a mild floral scent which I'm not in love with, but for a GREAT face wash I'm willing to deal with the fragrance.

Review 3

One of my Holy Grail products!! This may be my absolute favorite gentle exfoliating treatment ever!! Over the past 3 years, I've gone through 5 bottles. I can't stress enough how much this product changed my skin. First use, a slight tingling sensation but nothing uncomfortable and quickly went away. Immediately, this cleared my texture (gentle, not harsh on the skin). My skin looked plump and fresh, clear and smooth - a product that I can go back to when my skin is weird and it always works! I exfoliate a few times a week. I use it after cleansing, and before moisturizing. Lactic acid (an Alpha Hydroxy Acid) is more gentle than glycolic acid (LA is a bigger molecule), and a humectant (hydrating). Great for dry or sensitive skin. A slight sour milk smell, it is lactic acid after all, nothing unbearable. The pH isn't that high, so this product is great for someone who is just starting out with acids. There is some concern over the formaldehyde (dmdm hydantoin), to me it's not that bad. Small amounts aren't a huge concern for me. It helps the product last longer. The only downside is the PRICE (1.7 oz/\$158), so that's why the 4 stars. but for me it is absolutely worth it!!!

Review 4

I bought this to tint my sunscreen with and to use for extra bronzing in my makeup routine. Works pretty well and I also like that it's not shimmery. I find that it works best mixed in with a serum or lotion. When I dab and blend directly onto my skin (foundation or none) I will later notice it didn't blend all the way in.

Review 5

The salves tint gives your lips a hint of rosininess. I love how velvety the balm is and that you can use it not only on your lips but hands and to very dry facial skin. Although I feel like sometimes it is not moisturizing my lips the way it is my skin

RATING 5

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Review 1

I love this product! I was so excited to try it, and even though the smell isn't that great, the results are worth it for sure! Made my skin so moisturized and refreshed looking! Love it!

Review 2

I received this product from Influenster for free and I just have to try it and give my honest opinion. I ♥ this product, I use it before my make up, my skin feels and looks so good and fresh. I totally recommend. Thanks Influenster for the opportunity to try this product.

Review 3

This stuff is seriously liquid gold to me. I got a sample of it in my Play! box and had to buy a larger size. I used the sample for probably 2 weeks and still have some left, so the larger size will last you a loooong time. It comes with a little scoop that I use to get it out of the jar so I don't contaminate it. I use less than a dime size for my whole face and neck, and I feel like it's so hydrating for my combo skin but also doesn't break me out. Whenever I wear it, my skin looks and feels better in the morning, no doubt. Do yourself a favor and try it!

Review 4

This product is amazing!! Your skin feels so smooth. I have definitely noticed the difference under my eyes as well. They do t look as puffy. It makes your skin feel so clean and smells amazing!

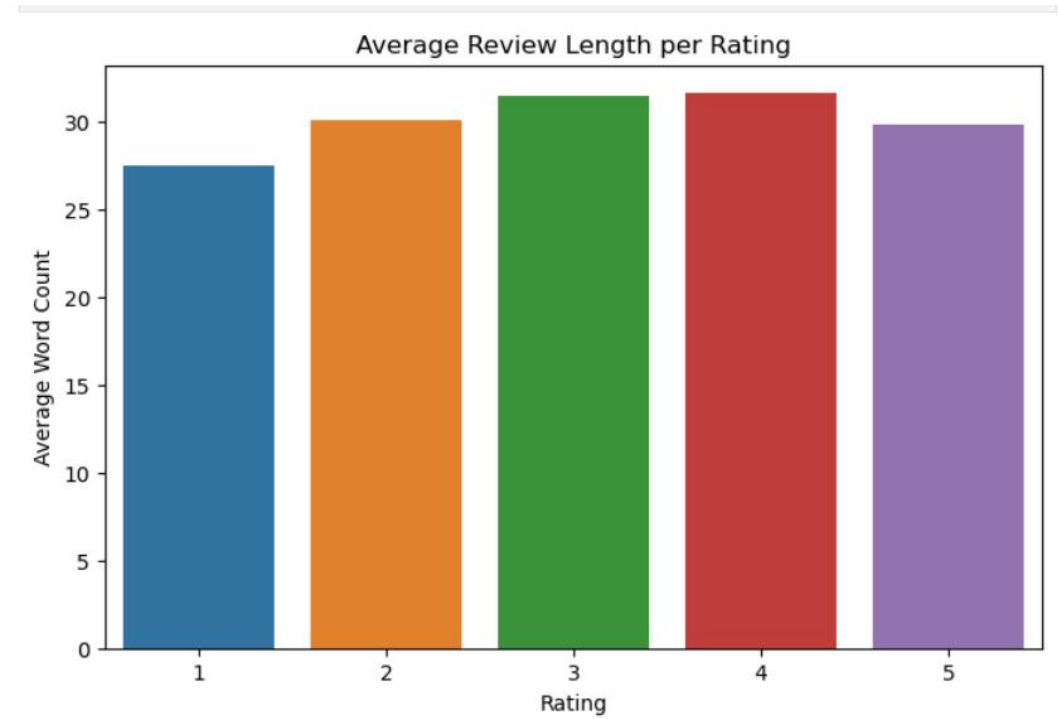
Review 5

I love this. I am always looking for a lotion that isn't over priced and makes my skin feel so soft. I just turned 30 so I am more prone to wrinkles now, especially with my dry skin. This lotion feels so good on my skin before I go to bed and it smells like cucumbers which makes me feel like I'm getting a professional facial. Will keep using long term!

Chart 5 - Average Review Length per Rating (Bar Chart)

sns.barplot showing mean word count per rating class:

Rating	Avg Word Count	Observation
★ 1	27.45	Shortest - concise negative feedback
★ 2	30.09	Slightly longer - more disappointment explanation
★ 3	31.48	Longest - balanced reviews need more words
★ 4	31.62	Similar to 3-star
★ 5	29.85	Shorter - simple positive statements



10. Balanced Dataset Creation

10.1 Strategy

Stratified random sampling without replacement: exactly 20,000 reviews were drawn from each star class using random_state=42 for reproducibility.

```
balanced_df = (  
    df.groupby('rating', group_keys=False)  
      .apply(lambda x: x.sample(n=20000, random_state=42))  
      .reset_index(drop=True)  
)
```

10.2 Result

Rating	Available	Sampled	Final %
★ 1	32,789	20,000	20.0%
★ 2	29,195	20,000	20.0%
★ 3	43,950	20,000	20.0%

★ 4	106,747	20,000	20.0%
★ 5	387,661	20,000	20.0%
Total	600,342	100,000	100%

10.3 Shuffle and Save

```
# Shuffle all 100K rows randomly
balanced_df = balanced_df.sample(frac=1,
random_state=42).reset_index(drop=True)

# Save to disk
balanced_df.to_csv('sephora_cleaned_dataset.csv', index=False)
# Output: Balanced dataset saved.
```

11. Stratified Train-Test Split

11.1 Defining X and y

```
X = balanced_df['clean_review'] # 100,000 cleaned review strings
y = balanced_df['rating']       # 100,000 integer labels (1-5)
```

11.2 Split Parameters

```
from sklearn.model_selection import train_test_split

X_train_text, X_test_text, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20, # 20% held out for testing
    random_state=42, # reproducible split
    stratify=y      # preserve class proportions in both sets
)
```

11.3 Results

Split	Total Rows	Per Class	Balance
Training (X_train_text, y_train)	80,000	16,000 each	Perfect - 20% per class
Test (X_test_text, y_test)	20,000	4,000 each	Perfect - 20% per class

Why stratify=y?

Without stratify, random splitting could produce slightly unequal class sizes by chance. stratify=y forces sklearn to maintain the exact 20%/20%/20%/20%/20% ratio in both splits, guaranteeing balanced evaluation.

12. TF-IDF Vectorization

12.1 What is TF-IDF?

TF-IDF (Term Frequency–Inverse Document Frequency) is a text vectorization technique used to convert textual data into numerical feature vectors that can be processed by machine learning algorithms.

In this project, TF-IDF is used to transform customer reviews into numerical representations, where each review is represented by the importance of the words it contains.

1. Term Frequency (TF)

Term Frequency measures how often a term appears in a specific document (review).

$TF(t,d) = \text{Number of times term } t \text{ appears in document } d \div \text{Total number of terms in document } d$

Where:

- **t** = term (word)
- **d** = document (review)

2. Inverse Document Frequency (IDF)

Inverse Document Frequency measures how rare or informative a term is across the entire dataset.

Formula:

$$\text{IDF}(t) = \log(N / \text{DF}(t))$$

Where:

- **N** = Total number of documents in the dataset
- **DF(t)** = Number of documents containing term *t*

Words that appear in many reviews receive lower IDF scores, while rare words receive higher IDF scores

3. TF-IDF Score

The TF-IDF score is calculated by multiplying Term Frequency and Inverse Document Frequency.

Formula:

$$\text{TF-IDF}(t,d) = \text{TF}(t,d) \times \text{IDF}(t)$$

A high TF-IDF score indicates that a word appears frequently in a particular review but is relatively rare across the entire dataset, making it a strong discriminative feature for machine learning.

12.2 Vectorizer Configuration

```
from sklearn.feature_extraction.text import TfidfVectorizer

tfidf = TfidfVectorizer(
    max_features=10000, # vocabulary capped at 10,000 terms
    ngram_range=(1, 2), # unigrams AND bigrams included
    min_df=5           # ignore terms in fewer than 5 documents
)
```

Parameter	Value	Effect
max_features	10,000	Caps vocabulary size. Each review becomes a 10,000-element sparse vector. Controls model complexity.
ngram_range	(1,2)	Captures both single words ('moisturize') and two-word phrases ('dry skin', 'breaks out'). Bigrams add context.
min_df	5	Discards terms in fewer than 5 reviews. Removes typos, rare codes, and non-generalizable terms.

12.3 Final Output Shapes

Variable	Shape	Type	Meaning
X_train	(80,000 x 10,000)	scipy sparse matrix	80K training reviews as TF-IDF vectors
X_test	(20,000 x 10,000)	scipy sparse matrix	20K test reviews, same vocabulary
y_train	(80,000,)	pandas Series (int)	Labels 1-5, 16,000 per class
y_test	(20,000,)	pandas Series (int)	Labels 1-5, 4,000 per class

Sparse Matrix	X_train and X_test are scipy sparse matrices - most of the 10,000 values per row are 0.0 (terms not in that review). Sparse format stores only non-zero values, saving enormous memory vs a dense 80,000 x 10,000 float array.
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